

Name:

Roll No.

Batch:

Signature :

MS101 – Makerspace
2023-24/II (Autumn Semester)
EE Quiz - 1

Jan 31, 2024 (Wed)

Time: 45 min

Marks: 30

1. This **Question-cum-Answer Booklet** has 4 pages and 7 questions.
2. Write your **answers only in the space provided for answers**. Answers written at any other place will not be checked. You may use the page margins for rough work.
3. No explanations / clarifications will be given to any of the questions.
4. No negative marks for wrong answers, however steps are required for all numerical answers.

1. A voltage source was made using two 12 V batteries connected in series, each having an internal resistance of 5 Ω .
 - A. What's the no-load voltage of the voltage source? (No-load voltage is defined as the voltage across the terminals of the voltage source when no load is connected, i.e. the load resistance is infinite).
 - B. If a load resistance of 100 Ω is connected across the terminals of the above voltage source, what will be the terminal voltage?
 - C. How much power in Watts will be dissipated in the 100 Ω load resistor?

No marks without steps.

Marks: 3

Steps:

Answer: A) V B) V C) W

2. A practical current source is modelled as an ideal current source of 200 mA in parallel with a resistor of 150 k Ω . When a load resistance R_L was connected across this source the current through R_L was found to be 125 mA. What is the value of R_L in k Ω ?

No marks without steps.

Marks: 2

Answer: R_L = ----- k Ω

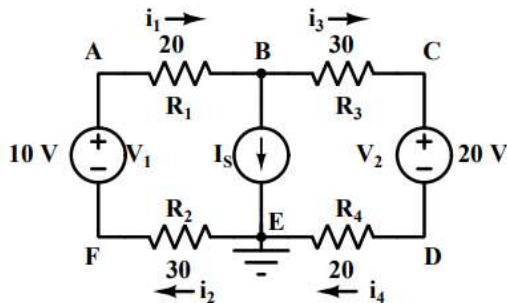
3. In the circuit shown below, the component values are:

$$V_1 = 10 \text{ V}, V_2 = 20 \text{ V}, I_S = 0.5 \text{ A}$$

$$R_1 = 20 \, \Omega, R_2 = 30 \, \Omega,$$

$$R_3 = 30 \, \Omega, R_4 = 20 \, \Omega.$$

Positive directions for currents are shown by arrows in the circuit diagram.



A) Find the values of i_1 , i_2 , i_3 and i_4 , and report these values in the table on the right. Calculation steps are to be shown in the space below or on the right side. (Hint: use superposition method):

Marks for 3A: $1 \times 4 = 4$

Answers for 3A:

Currents in mA (with sign)			
i_1	i_2	i_3	i_4

Marks for 3B: $1 \times 3 = 3$

Answers for 3B

Voltages at:		
A	B	C

B) Find the voltages at nodes A, B, and C in the circuit above and report these in the table on the right. Calculation steps are to be shown below or on the right side:

4. The current through a *pn* junction diode in the forward region can be approximated as: $i_d = I_S \cdot \exp [v/V_T]$, where v is the voltage across the diode, I_S is the reverse-saturation current and V_T is the thermal voltage. $V_T = 25 \text{ mV}$, and $I_S = 10^{-14} \text{ A}$.

A resistor and the above *pn* junction diode are connected in series and then the series combination is connected across a battery of 5 V such that the diode is forward biased. If the voltage across the resistor is 4.3 V calculate the diode current i_d in mA. Your current should be precise to the first decimal place.

Marks: 2

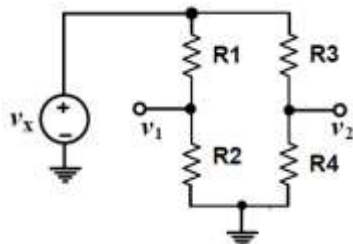
Show steps here.

Answer: $i_d =$ mA

5. A Wheatstone bridge is shown below where v_x is the bridge excitation voltage and v_1 and v_2 are the output terminal voltages. Assume the differential-mode (DM) voltage of the bridge to be: $v_{id} = v_1 - v_2$.

Given: $v_x = 10$ V, $R_1 = R_2 = R_4 = 10$ k Ω , $R_3 = 9.5$ k Ω .

- What is the DM voltage of the above bridge in millivolts?
- What is the common-mode (CM) voltage of the above bridge in volts?
- If the output terminals v_1 and v_2 were connected to an op amp with a differential gain of $A_d = 20$ and a common-mode gain of $A_c = 0.01$, what would be the output voltage of the op amp?



Marks: 6 (= 2 + 2 + 2)

Show steps here for A and B. *No marks without steps.*

A) DM voltage = mV

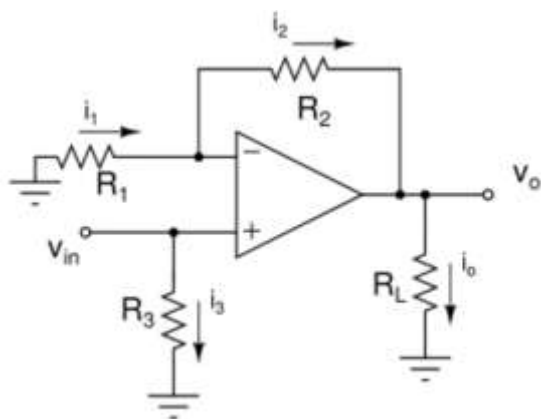
B) CM voltage = V

Show steps here for C

C) Output voltage = V

6. In the op-amp circuit shown in the figure, $R_1 = 150$ k Ω , $R_2 = 75$ k Ω , $R_3 = 1$ M Ω , and $R_L = 1$ k Ω .

Find the voltage, current, and power gains.



Marks: 5 (= 1.5+2+1.5)

Steps must be shown. *No marks without steps.*

A) Voltage gain $A_v = v_o / v_{in}$

Steps:

Answer:

B) Current gain $A_i = i_o / i_3$

Steps:

Answer:

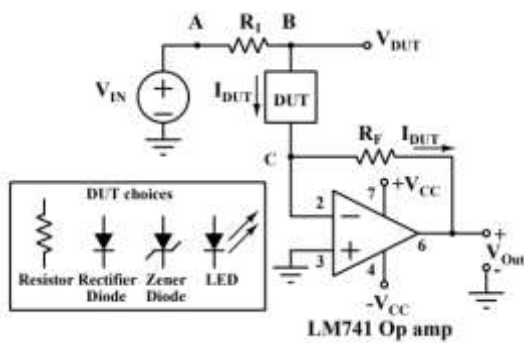
C) Power gain $A_p = A_v A_i$

Steps:

7. Circuit diagram of the op amp I to V converter used in Expt. 2 is shown below. The device under test (DUT) is connected between terminals B and C. Assume that the op amp is ideal. Also assume that the maximum and minimum V_{Out} levels are $+V_{CC}$ and $-V_{CC}$ respectively

$+V_{CC} = +12\text{ V}$, $-V_{CC} = -12\text{ V}$; $R_1 = 1\text{ k}\Omega$, $R_F = 2\text{ k}\Omega$.

- A) A $3\text{ k}\Omega$ resistor is connected as the DUT. If $V_{IN} = +6\text{ V}$, what will be the values of V_{DUT} and V_{Out} ?
- B) A Zener diode is connected as the DUT with its anode at terminal B and cathode at terminal C. The Zener voltage is 6.7 V , and the Zener I-V characteristic is linear in the Zener region (avalanche breakdown region) with a slope of $150\text{ }\Omega$. If $V_{IN} = -9\text{ V}$, what will be the values of V_{DUT} and I_{DUT} ?



Marks: 5 (= 2 + 3)

A) Show steps here

Answer: $V_{DUT} = \dots\dots\dots\text{ V}$, $V_{Out} = \dots\dots\dots\text{ V}$

B) Show steps here

Answer: $V_{DUT} = \dots\dots\dots\text{ V}$, $I_{DUT} = \dots\dots\dots\text{ mA}$